

Nutritional Profiles in USDA Foundation Foods: Implications for Plant-Based Home Cooking

Author: Thomas Lu

Date: February 2026

Abstract

This study compares the nutritional profile between animal-based and plant-based “Foundation Foods¹” to identify practical dietary implications for home-based plant-centered diets. A comparative statistical analysis was performed on 215 unique USDA Foundation Food items (n=80 animal-based; n=135 plant-based). Nutrients were standardized per 100g and 100kcal to evaluate macronutrient distribution, micronutrient efficiency² (Iron, Calcium), and Vitamin B12 prevalence. Animal-based foods demonstrated significantly higher median protein (16.6g vs. 2.41g) and caloric density (156.5kcal vs. 44.0kcal), driven by a 44% fat-to-calorie ratio. Conversely, plant-based foods were primarily carbohydrate-driven (59% of calories) and exhibited a 10 times higher median Iron density per 100 kcal (1.18mg vs. 0.11mg). Vitamin B12 was absent in 98.6% of plant-based samples, appearing only in fortified soy products. The data shows that to achieve high protein in plant-based home cooking requires either a significant increase in total food volume or a prioritized focus on high-protein outliers (e.g., legumes, soy). While plant-based whole foods offer superior mineral efficiency and fiber, the lack of B12 and lower protein-to-weight ratios requires strategic fortification and portion planning to reach nutritional needs.

Introduction

In modern dietetics, plant-based and animal-based have been the center for comparative nutritional analysis, with the majority of evaluations centered around processed substitutes. This research shifts the focus to the USDA Foundation Foods, which are the unprocessed or minimally processed ingredients used for home cooking. Examples include whole grains, legumes, raw meats, and fresh produce. When switching to a plant-based diet, the implications involve more than the substitution of protein sources, requiring a significant change in the structure and volume of the meal. Plant-based whole foods are characterized by a unique nutrient profile with the benefits of dietary fiber and a broad variety of carbohydrates. However, the benefits come with lower protein-to-weight ratio and the absence of Vitamin B12, among other nutrients.

¹ Categorized by the USDA, refers to unprocessed or minimally processed foods

² Nutrient efficiency refers to the mass of nutrient in a 100 kcal sample

By standardizing these unprocessed ingredients on a gravimetric (100g) and metabolic (100 kcal) scale, this study investigates the physiological benefits trade-offs associated with plant-based whole foods. The findings provide an evidence-based framework to manage portion sizes, maximize mineral efficiency, and identify critical fortification requirements, ensuring the plant-based diets are both calorically sustainable and nutritionally complete.

Methods

1. Data Selection

To model a “home-cooking” dietary environment, the analysis was restricted to unprocessed or minimally processed foods. Examples include raw meats and fish (e.g., Atlantic salmon), legumes (e.g., dry beans), vegetables (e.g., raw kale), eggs, and plain dairy products.

2. Data Acquisition

A total of 215 unique food items were sourced from the USDA FoodData Central database. For all items, nutrient values per 100g were used to standardize comparisons across food categories.

3. Comparison Groups

Foods were classified into two groups for comparative analysis:

- Animal-based (n=80): meats, fish, eggs, and plain dairy products.
- Plant-based (n=135): raw legumes, nuts, seeds, and vegetables.

4. Data Processing

The raw nutrient dataset was cleaned and standardized. Nutrient fields were subset and renamed to ensure consistency. Related fatty-acid, carbohydrate, and fiber components were integrated into total fat, carbohydrate, and dietary fiber measures. Energy values reported in kilojoules were converted to kilocalories (1 kJ = 0.239 kcal). When caloric values were missing, energy was calculated using Atwater factors³. Logical corrections were applied to resolve inconsistencies, such as cases where total carbohydrates were lower than total dietary fiber, negative carbohydrate values, and missing nutrients. The final analytic dataset included the following standardized variables: Food, FoodType, FoodCategory, Calories, Protein, Fat, Carbs, Fiber, Calcium, Iron, and Vitamin B12.

³ 4 kcal/g for protein and carbohydrate; 9 kcal/g for fat

5. Data Analysis

All processing and statistical analyses were performed in Python 3.x. Pandas was used for data manipulation, and Seaborn and Matplotlib were used for visualization.

Results

Statistical Summaries

1. Macronutrient Distribution

Animal-based calories are derived almost exclusively from Protein (46.06%) and Fat (43.77%). In contrast, plant-based sources are primarily carbohydrate-driven (58.93%), though they remain a significant source of protein (26.77% of total caloric intake).

Table 1: Macronutrient to calories ratios

Food Type	Protein Percentage	Fat Percentage	Carbs Percentage
Animal-based	46.06%	43.77%	9.49%
Plant-based	26.77% ⁴	15.59%	58.93%

Note: Relative macronutrient ratios do not reflect absolute nutrient density

2. Micronutrient Efficiency

When standardizing for caloric intake, plant-based foods proved more efficient for Iron delivery, with a median density of 1.18 mg per 100kcal compared to just 0.11 mg in animal products. Although plant-based foods contain more calcium per 100 kcal (median 53.70 mg vs. 27.98 mg)⁴, animal-based foods provide substantially more calcium per 100 g (mean 188.50 mg vs. 64.74 mg).

Table 2: Nutrient density per 100 kcal

Food Type	Protein (g)		Iron (mg)		Calcium (mg)		Vitamin B12 (µg)	
	mean	median	mean	median	mean	median	mean	median
Animal-based	11.52	8.78	0.30	0.11	89.88	27.98	0.79	0.32
Plant-based	6.69	5.11	1.86	1.18	104.20	53.70	0.03	0.00

3. The Vitamin B12 Gap

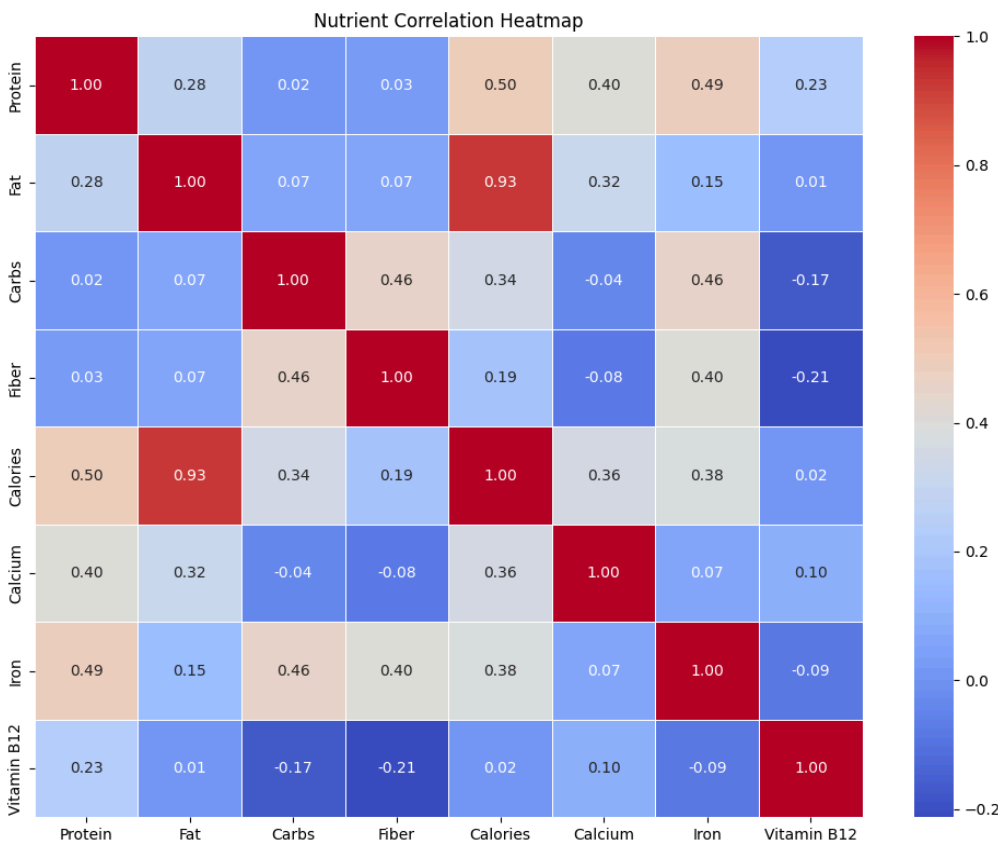
Vitamin B12 showed a near-total absence in the plant-based group. The presence rate for B12 in animal-based foods was 58.7%. Within the plant-based group, only fortified soy products (1.4% of the group) contained measurable amounts of the vitamin.

⁴ Note that leafy greens are very low in calories

4. Correlation Analysis

Protein has a moderate positive correlation with Calories (0.50) and Iron (0.49), suggesting that high-protein sources in this dataset are frequently accompanied by iron, regardless of the food type. This reflects the co-occurrence of these nutrients in dense, high-protein foods (e.g., red meats, legumes), rather than implying a causal relationship. Carbs and Fiber have almost zero correlation with Protein in this dataset (<0.02), indicating no meaningful linear association in this dataset.

Figure 1: Nutrient Correlation Heatmap⁵



Graphs

1. Average Macronutrients

Macronutrients: Animal-based foods lead significantly in average Protein (16.8g vs 7.6g) and Fat (14.2g vs 5.9g). Plant-based foods dominate in Carbs (12.1g vs 2.8g) and Fiber (3.6g vs approx. 0g).

⁵ Correlations were generated using Seaborn's correlation heat map, which computes Pearson correlation coefficients; and were computed across the full dataset (n=215). Correlation coefficients therefore reflect linear associations across the entire dataset rather than within specific food categories.

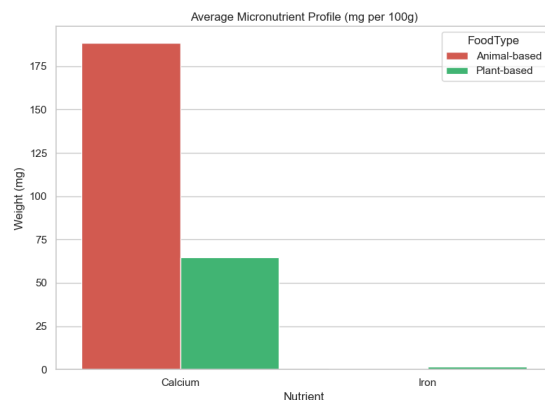
Graph 2: Average macronutrient per 100g



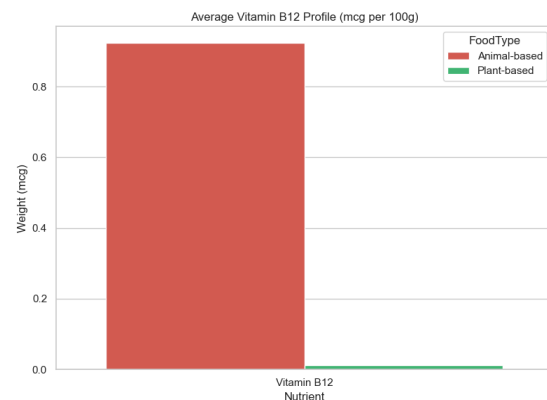
2. Average Micronutrient

Micronutrients (mg): Animal-based foods show a higher average Calcium (188.5mg vs 64.7mg), whereas plant-based foods provide nearly 3 times the Iron on average (1.85mg vs 0.66mg). For Vitamin B12, animal products average 0.92µg, while plant products are near-zero (0.01µg), primarily appearing only in fortified soy milk.

Graph 3: Average micronutrient per 100g



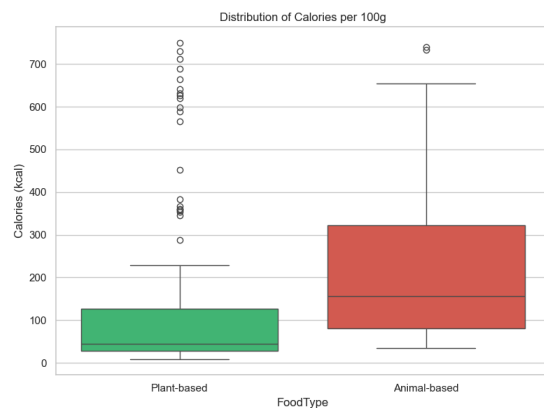
Graph 4: Average Vitamin B12 per 100g



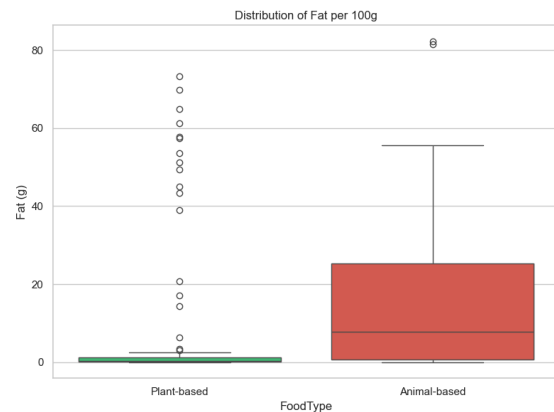
4. Calories Distribution

Animal-based sources are significantly more calorie-dense (median = 156.5 kcal vs 44.0 kcal). This disparity is primarily caused by fat distribution, where animal products have a median nearly twenty times higher than plant products (median = 7.72g vs 0.40g). Although both show high variability (SD $\approx$ 16g) due to caloric-dense outliers like fatty meats, butter, and oil-rich seeds, the majority of plant foods remain low-fat (Q3 = 1.28g).

Graph 5: Average calories per 100g



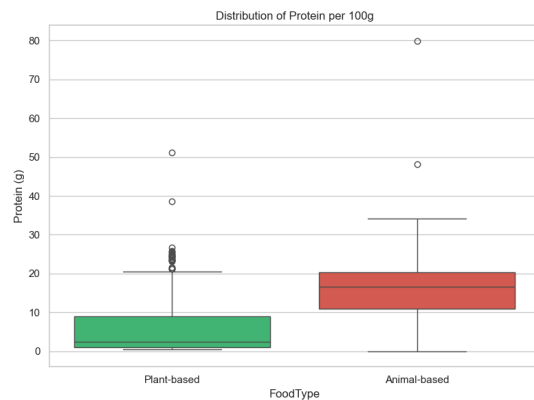
Graph 6: Fat distribution per 100g



5. Protein Distribution

Animal-based sources yield a median protein content nearly seven times greater than plant-based sources (16.60g vs 2.41g). While animal products maintain a consistently high protein baseline (Q1 = 10.93g), most plant-based ingredients are low in protein (Q3 = 9.04g). Most plant-based proteins are found in outliers, such as legumes and soy products (Max = 51.1g).

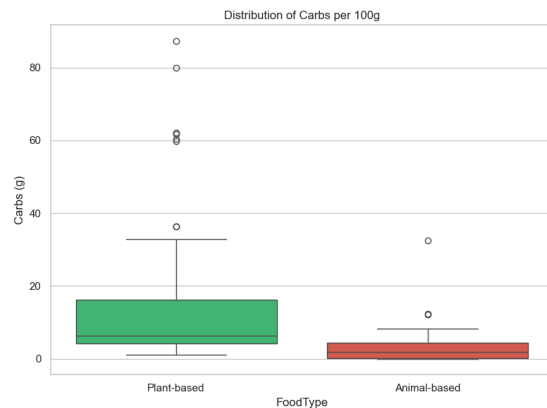
Graph 7: Protein distribution per 100g



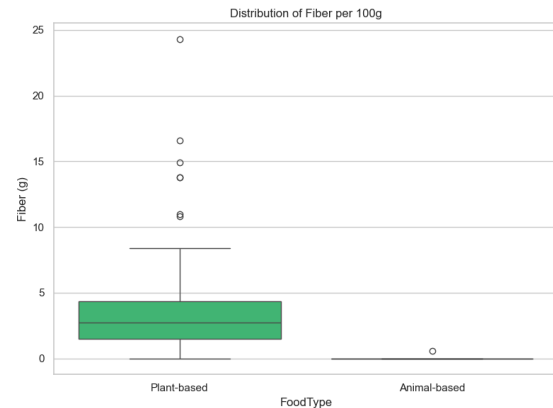
5. Carbohydrates Distribution

Animal-based foods show minimal carbohydrate content (median = 1.89g) and a total absence of dietary fiber (median = 0.0g, only outlier being strawberry yogurt). In contrast, plant-based foods have a diverse carbohydrate profile (mean = 12.14g; max = 87.3g) that correlates with dietary fiber (mean = 3.56g).

Graph 8: Carbs distribution per 100g



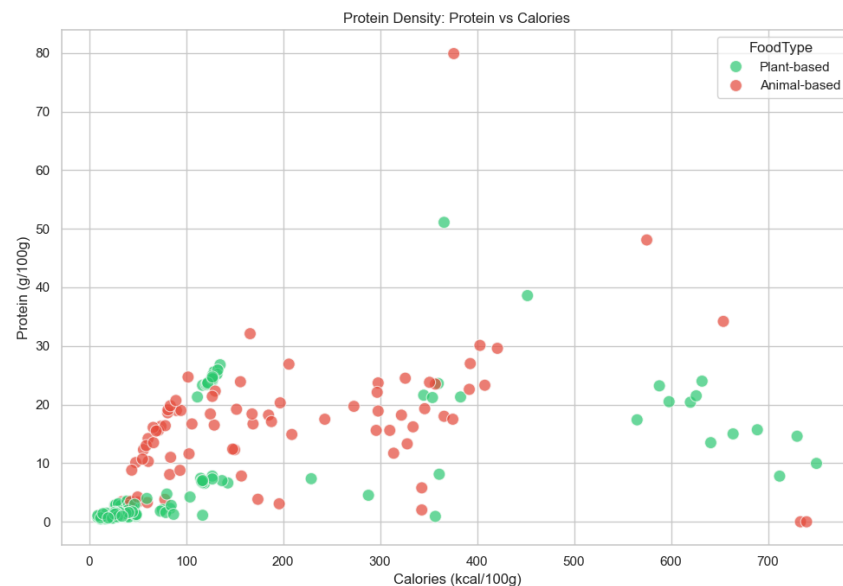
Graph 9: Fiber distribution per 100g



6. Protein Efficiency and Density

There is a moderate positive correlation ($r=0.50$) between caloric density and protein content, though it varies significantly by food type. Animal-based foods have higher protein concentration (median = 16.6g vs 2.41g). The upper-left section with maximum efficiency ($>24\text{g}$ protein per 100 kcal) mostly consists of wild-caught fish. Plant-based sources are more dispersed but have high-efficiency outliers in the legume products ($\sim 20\text{g}$ of protein per 100 kcal)

Graph 10: Protein Density (Protein vs. Calories)



Discussion

The results of this analysis shows a divergence in the nutritional profile between plant-based and animal-based USDA Foundation Foods. Animal-based foods are concentrated in protein and lipids, with 90% of their caloric profile derived from these two macros (Table 1). Conversely, plant-based foods are high-volume, carbohydrate sources that provide dietary fiber (Graph 9). Achieving high protein through plant-based whole foods requires either:

- a significant shift in plate volume; for instance, to match 20g of chicken breast in protein, 220g of kale or 90g of chickpeas is needed⁶.
- prioritizing high-protein outliers⁷; specific outliers like soy flour (51.1g), 0% moisture tan beans (26.8g), dry-roasted almonds (20.4g), and other legume products can match animal protein.

For micronutrients, the data suggests that plant-based foods are remarkably efficient at delivering Iron per calorie⁸, with ten times the median density of animal products (Table 2). However, vitamin B12 has a 0.00µg median in plants (Graph 4). Home cooks transitioning to plant-based diets must intentionally incorporate the small subset of fortified items (1.4%) or external supplementation to avoid deficiency. Furthermore, while animal foods appear superior in Calcium per 100g (Graph 3), the higher Calcium density per 100kcal in plants (Table 2) indicates that leafy greens and legumes are more efficient sources than dairy for calorie-conscious consumption⁸.

Several limitations within this study must be considered when applying these findings to real-world cooking. The plant-based category (n=135) contains a high proportion of raw vegetables which are predominantly water. This depresses median protein, fat, and calorie values while inflating their nutrient-per-100kcal stats. Likewise, nuts and seeds tend to be lower in protein relative to their energy content, which reduces the overall mean protein density of the plant-based group. The larger sample size of plant-based foods (135) versus animal-based (80) may introduce skew due to a larger amount of low-nutrient outliers that lower the group's median protein and calorie values. Statistics are standardized to 100g, which cannot be directly applied to a culinary setting. Additionally, the dataset accounts for nutrient content and not absorption.

Future studies should integrate bioavailability and adjust total nutrient values to “absorbed” values. Finally, the dataset can be expanded to include other factors (such as Vitamin C, fat-soluble vitamins, essential amino acids) which would provide a more holistic view of nutrients absorption and physiology.

⁶ Values are approximated based on USDA data

⁷ While not in the present analysis, it is important to note that most plant-based protein sources do not contain all essential amino acids in adequate proportions. Plant proteins require combining complementary sources (e.g., legumes with grains) to achieve a complete amino acid profile.

⁸ Note the bioavailability of non-heme iron and calcium with oxalates.

Conclusion

This study investigates how the nutritional profiles of unprocessed plant-based and animal-based USDA Foundation Foods change the practicality of plant-based home cooking. The findings demonstrate a clear divergence in nutrient density: animal-based foods provide substantially higher protein and calories, while plant-based foods deliver superior iron and calcium efficiency per calorie along with dietary fiber. The differences suggest that a whole-food plant-based diet requires deliberate adjustments to meal volume, ingredient selection, and nutrient planning. In particular, achieving protein and Vitamin B12 intake requirements demand prioritizing high-protein plant outliers (such as legumes and soy products) and incorporating fortified foods or supplementation due to their lower protein-to-weight ratios and absence of vitamin B12. This analysis reinforces that nutritionally complete plant-based home cooking is achievable but must be approached with intentional portion and nutrient planning rather than simple substitution of animal foods.

Data Availability

The full cleaned dataset used in this analysis is available in Google Sheets at:

 [Project - Nutritional Profiles in USDA Foundation Foods, Full Data & Useful Stats](#) .

The original USDA raw nutrient files and all analysis code (Python scripts) are available in the GitHub repository at: [tlu0044/plant-animal-nutrition-comparative-analysis](https://github.com/tlu0044/plant-animal-nutrition-comparative-analysis).